DFT + SCAPS-1D / LEAD-FREE PEROVSKITE SOLAR CELLS

Unveiling the Photovoltaic Potential of $CsSnF_3$

Insights from first-principles theory and numerical simulation studies

Mohammad Borhan Uddin[a] • **Asif Noor Jamee**[b] • **Siratul Mostakim**[b] • **Salman Farsi**[a] • **Ashir Hasan Nibir**

[a] Materials Science & Engineering, KUET | [b] Electrical & Electronic Engineering, East Delta University

Unpublished draft, not yet submitted for peer review

Crystal Structure & Electronic Structure

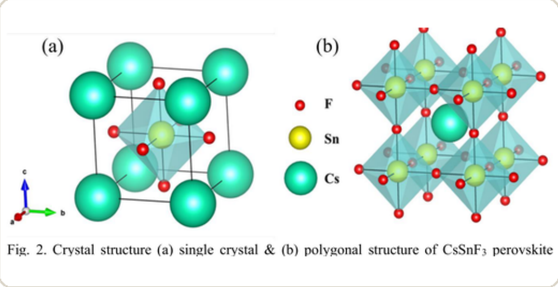


Fig. 2. Crystal structure (a) single crystal & (b) polygonal structure of $CsSnF_3$ perovskite

Fig. 2 · $CsSnF_3$ manuscript · PDF p.12

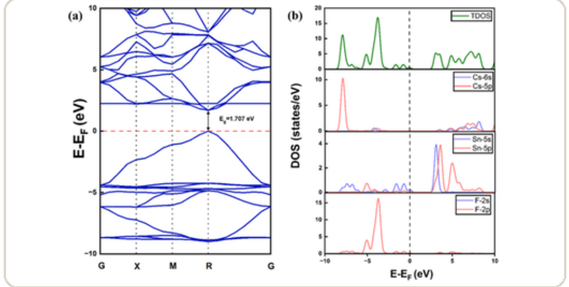


Fig. 3 · $CsSnF_3$ manuscript · PDF p.13

Cubic Pm-3m (No. 221); a = 4.794 Å; formation energy = −5.5 eV.

DFT-calculated direct bandgap = 1.707 eV at R. GGA-PBE likely underestimates the gap; hybrid-functional literature ≈1.9–2.1 eV.

Conduction band: mainly Sn-5s/5p; valence band: mainly F-2p.

Charge Density & Chemical Bonding

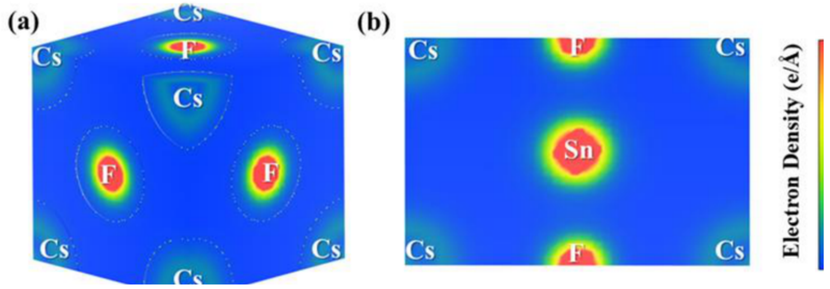


Fig. 4 · $CsSnF_3$ manuscript · PDF p.15

Charge density is localized around F atoms. The manuscript describes Cs–F as ionic and Sn–F as partly covalent.

Methods: CASTEP · GGA-PBE · OTFG ultrasoft pseudopotentials · 500 eV cutoff · Monkhorst–Pack k-point sampling.

Optical Properties

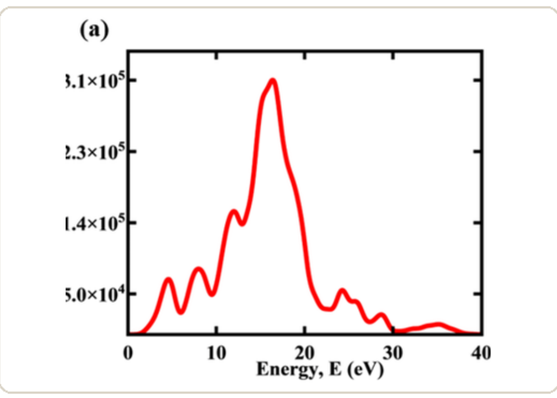


Fig. 5(a) · $CsSnF_3$ manuscript · PDF p.16

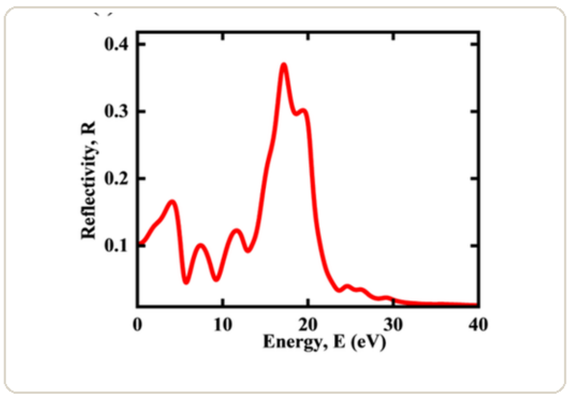


Fig. 5(f) · $CsSnF_3$ manuscript · PDF p.16

DFT-calculated absorption coefficient approaches 10^5 cm^{-1}; visible-region reflectivity is below 15%.

Figures from the manuscript; optical curves are DFT-calculated.

Device Architecture & Simulated Optimization

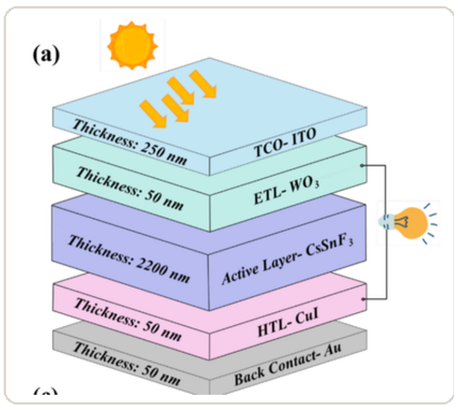


Fig. 1(a) · PDF p.10. Initial thicknesses; optimised values at right. Figure from the manuscript.

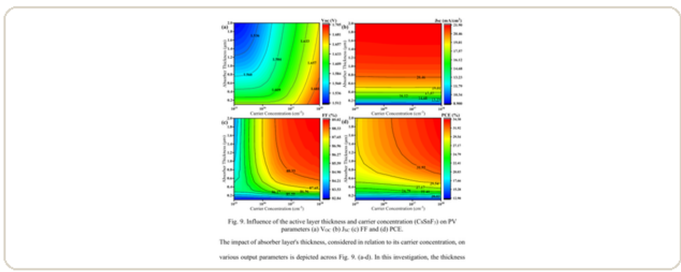


Fig. 9 · PDF p.24. Figures from the manuscript.

Optimised simulated device: PCE 30.72%; Voc 1.62 V; Jsc 21.94 mA cm^{-2}; FF 86.37%.
Optimised parameters: absorber 1.2 µm; carrier concentration 1.0×10^{18} cm^{-3}; absorber bulk defects 1.0×10^{14} cm^{-3}; WO_3 ETL 0.035 µm; CuI HTL 0.05 µm; interface defect density $<1.0\times10^{10}$ cm^{-2}; Au work function 5.1 eV.

RESULT STATUS All device-performance values are SCAPS-1D simulated; material properties are DFT-calculated.